

- Q.24 Write are the conditions for parallel operation of an alternator? (CO3)
- Q.25 Explain voltage regulation of an alternator. (CO3)
- Q.26 Write advantages of squirrel cage induction motor. (CO1)
- Q.27 Explain working of reluctance Motor. (CO2)
- Q.28 Define cogging and crawling in 3D motor. (CO2)
- Q.29 Draw and explain V-curves of a synchronous machine. (CO3)
- Q.30 Discuss working of AC servo motor. (CO4)
- Q.31 Explain why the stator core of an alternator is laminated. (CO3)
- Q.32 Explain operating principle of 3d induction motor. (CO4)
- Q.33 Write down various applications of synchronous motors. (CO2)
- Q.34 Discuss the cause of hunting and its preventive methods. (CO1)
- Q.35 Write a short note on the armature reaction of a synchronous machine. (CO3)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the working principle & construction of 3 phase synchronous machines. (CO3)
- Q.37 Draw and explain the torque-slip characteristics of 3 phase induction motors. (CO1)
- Q.38 Describe construction and working of Hysteresis motor with its applications. (CO2)

No. of Printed Pages : 4

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5th Sem /

Elect, Power Station Engg., Elect & Eltx. Engg.

Sub : Electrical Machines-II

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What is the application of synchronous condensor to (CO3)
- Control of active power
 - Control of reactive power
 - Control of apparent power
 - None of above
- Q.2 The inverted V plots is plotted between ____ (CO3)
- Power factor vs field current for constant shaft load
 - Field current vs power factor for variable shaft load
 - Armature current vs terminal voltage
 - terminal voltage vs power factor
- Q.3 The condition which allows the torques to be taken care for choosing a synchronous motor is/are?(CO3)
- Starting torque
 - Pull in torque
 - Pull out torque
 - All of the mentioned

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- Q.4 Which among the following are the applications of synchronous motors? (CO3)
- Synchronous condensor
 - Blowers
 - Fans
 - Compressors, fans, blowers
- Q.5 The frequency of voltage generated in large alternators in india is (CO1)
- 0 Hz
 - 230 Hz
 - 60 Hz
 - 50 Hz
- Q.6 The main function of a starter is (CO5)
- To start the motor
 - To stop the motor
 - To limit the starting current
 - To limit the applied voltage
- Q.7 Slip rings of phase wound induction motor are made of (CO2)
- Wood
 - Cast iron
 - Steel
 - Copper
- Q.8 which motor is generally used in tape recorders (CO4)
- Hysteresis motor
 - Split Phase motor
 - Reluctance motor
 - Universal motor
- Q.9 In a motor, which loss varies with load (CO4)
- Core loss
 - Copper loss
 - Bearing friction loss
 - Windage loss

- Q.10 Which type of motor is used in ceiling fan (CO2)
- Shade pole
 - Universal motor
 - Permanent capacitor start
 - Capacitor start capacitor run

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is synchronous speed? (CO1)
- Q.12 Define generator. (CO1)
- Q.13 Give any two applications of slip ring induction motors. (CO4)
- Q.14 Give expression for % slip? (CO3)
- Q.15 Write down the full form of LIM? (CO4)
- Q.16 Define voltage Regulation? (CO1)
- Q.17 Synchronous motor runs at _____ speed. (CO1)
- Q.18 Define stepper motor. (CO4)
- Q.19 Define coil span factor. (CO3)
- Q.20 Over excited synchronous motor behaves like a _____. (CO3)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain anyone starting method of the 3 phase induction motor. (CO1)
- Q.22 Compare Squirrel cage and slip ring induction motor. (CO3)
- Q.23 Explain the methods of cooling of a synchronous machine. (CO3)